



Confuciusornis fossil

Lots of preserved *Confuciusornis* specimens have long tail streamers. These are now thought to be exclusive to males and to have been used in displays to attract a mate during breeding season.

2 Archaeopteryx

Thought to be the first true bird, *Archaeopteryx* had feathered wings, but retained dinosaur characteristics, such as clawed forelimbs, a toothed beak, and a bony tail. Its small wing muscles suggest it may not have been able to fly well.

Claws on the bird's thumb and third finger may have helped it climb through trees.

Mass extinction

Life in the prehistoric past was occasionally rocked by catastrophic events that wiped out entire groups of organisms. Five mass extinction events have occurred in the last 500 million years. Many may have been caused by climate change and volcanic eruptions, but there is also strong evidence that the event that eliminated the dinosaurs was caused by an asteroid striking the Earth.



3 Confuciusornis

Thirty million years after *Archaeopteryx*, *Confuciusornis* emerged. It had a tail of feathers that lacked a bony support, and a toothless beak. Its flight feathers were longer than *Archaeopteryx*, but it still could not flap as well as modern birds.

4 Ichthyornis

Living just before the extinction of the dinosaurs, *Ichthyornis* resembled a modern seabird, and was about the size of a gull. Although it had strong flight muscles, its bill still contained sharp teeth that helped it catch fish.

Ichthyornis had a well-developed breastbone for supporting strong flight muscles.

A mass extinction 66 million years ago drove dinosaurs to extinction, but their bird descendants survived.

5 Ruby-throated hummingbird

A lightweight skeleton and strong muscles help most modern, toothless birds fly with far greater skill than any of their ancestors.

